THE FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF MATHEMATICS AND STATISTICS
1ST AND 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

Programme: HND I **Course Code:** STA314 **Course Title:** Operations Research
Instructions: Answer any Five (5) Questions; Time Allowed: 3 Hours

1. (a) Bature is a livestock farmer and needs to mix feed from corn and millet. The cost of each of these grains differs and each grain has different amount of vitamins. These costs and vitamins quantities are as follows:

Vitamin	Corn	Millet	Minimum Daily Requirement
A	3	5	15
C	8	8	32
E	7	3	21
Costs(N)	7	9	

Formulate the Linear Programing Problem, if Mustapha wishes to find the least mix that satisfies the vitamin requirement?
(b) State the generalized form of the Linear Programming Problem?

2. (a) Solve geometrically, the linear programming problem:

$$Max\ Z = 3x_1 + 4x_2$$
subject to
$$4x_1 + 2x_2 \le 100$$
$$4x_1 + 8x_2 \le 180$$
$$x_1 + x_2 \le 40$$
$$x_1 \le 20$$
$$x_2 \ge 10$$
$$x_1,\ x_2 \ge 0$$

b (I) What is a model (ii) State the principles of modeling in OR

3. Solve the Linear Programming Problem using the simple method:

$$Max\ Z = 3x_1 + 2x_2$$
subject to
$$x_1 + 2x_2 \le 6$$
$$2x_1 + x_2 \le 8$$
$$-x_1 + x_2 \le 1$$
$$x_2 \le 2$$
$$x_1,\ x_2 \ge 0$$

Arrow series

4. (a) Discuss briefly the nature of Operations Research
   (b) Discuss briefly any three (3) distinguishable characteristics of Operations Research

5. (a) Defined Normative and Descriptive models
   (b) Write short notes on any five (5) of the following: Slack of an activity, Slack of an event, Critical path, Replacement model, Queuing model, decision model, Competitive and Allocation models.

6. The following information shows the activities that must be performed in order to produce a new branded Loya powdered Milk by Promaxidor Nigeria Plc:

Activity	A	B	C	D	E	F	G	H	I	J	K
Preceding Activity	-	-	-	A	B	B	B	C	D,E	F,H	G
Duration(Days)	4	9	10	5	8	12	6	8	4	7	8

(a) Construct the network diagram of the above information
(b) Find the length of each path
(c) Find the critical path
(d) What is the expected duration for the completion of the project.

7. Consider the following project network. Using PERT, suppose that the usual three estimates for the time required (in months) for each of these activities are :

Activity	a	m	b
(1,2)	7	8	9
(1,3)	5	7	8
(2,6)	6	9	12
(3,4)	4	4	4
(3,5)	7	8	10
(3,6)	10	13	19
(4,5)	3	4	6
(5,6)	4	5	7
(5,7)	7	9	11
(6,7)	3	4	8

Designating the start of the roject as time 0, the scheduled time by contra t is 25 months,
(a) on the basis of the abov estimates, calculate the expected value and sta dard deviation of the time required for each acti ity.
(b) Using expected times determine the critical path for the project

THE FEDERAL POLYTECHNIC, MUBI

SCHOOL OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
HND First Semester Examinations 2021/2022 Academic Session

Programme: HND I CS   Course Code: COM 313
Course Title: **Computer Programming Using C++**   CU: 4.0
Instructions: *Answer any Four (4) Questions; Total marks 60; Time Allowed: 2½hrs Sept. 2022*

Question 1.

a) Define C++. **[4 Marks]**
b) When do we say that a Language is statically typed? **[5 Marks]**
c) Write the Basic Skeleton of C++ Program **[6 Marks]**

Question 2.

a) List the three data types in C++. **[4 ½ Marks]**
b) Declare a variables **Fees** and **Gender** that would hold value **94.99** and **Male** respectively? **[6 Marks]**

c) Predict the output of the following: **[4 ½ Marks]**

```
int x = 10;
x += 5;
x -=2;
++x;
cout << "x = " << x;
```

20

Question 3.

a) Is it possible to **switch case** with a **double data type** variable in **C++**? **[4 Marks]**
b) What will be the output of the following code segment? **[5 Marks]**

```
int i = 10, j = 5, k = 20;
int p = !(i>j && k > i);
cout <<" p is "<<p<< endl;
```

P = 0

c) What is the output of the following code segment: **[6 Marks?]**

```
int x = 5;
switch(x){
    default:
        x = x * 2;
        case 2:
        x = x + 10;
        break;
    case 5:
        x = x - 5;
    case 6:
        x = x + 1;
        break;
}
cout << "x = " << x;
```

output ?

Question 4.

a) Debug the following C++ program by stating the line numbers where errors are, and state the output **[7 Marks]**

```
1. #includ <iostream>
2. using namespace std;
3. int main(){
```

```
int aNumber = 100;
int bNumber = 200
int cNumber = 300;
if (aNumber <= bNumber && !(cNumber>bNumber)){
cout < aNumber + bNumber << endl;
}
els{
cout << aNumber - bNumber << endl;
}
return 0;
```

b) Write a C++ program using Switch Case to display days of a week (1-7 for Monday – Friday e.g if user select 3 it should be Wednesday). **[8 Marks]**

Question 5.

a) What will be the output of the following program? **[5 Marks]**

```
#include <iostream>
using namespace std;
int main(){
    char ch = '5';
    if (ch >= 'a' && ch<= 'z'){
        cout << "Hello, World!" << endl;
    }
    else if(ch >= '0' && ch <= '9'){
        cout << "Hello, Mars!" << endl;
    }
    else {
        cout << "Hello, Moon!" << endl;
    }
    return 0;
}
```

b) Using **While loop** write a C++ Program to display the even numbers from 2 to 16. **[6 Marks]**

C) Write the syntax of Switch case Statement. [4 Marks]

Question 6.

a) How many times the following loop will iterate? **[6 Marks]**

```
int i = 100;
while(i <= 200);
{
cout << "Testing" << endl;
++i;
}
```

b) Using and if-else statement write a program in C++ to find the largess of three numbers stores in 3 variables **x,y,z**. **[5 Marks]**

c) Given the following array definition answer the question that follows: **[4 Marks]**

```
using namespace std;
int main()
{string grading[5]={"Distinction","Upper Credit","Lower
Credit","Pass"};
double gpa[5]={4.5,3.5,2.5,2.0};
    cout << grading[3]<< endl;
    cout << gpa[1] <<endl;
    cout << gpa[4] <<endl;
    return 0;
}
```

i) What is the Data in **grading** [3] and **gpa**[1] ?
ii) Write a line of code that would replace Distinction in grading with **First Class**
iii) What would be the content of **gpa**[4]

THE FEDERAL POLYTECHNIC, MUBI
SCHOOL OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
END OF 1ST AND 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

PROGRAMME: HND I COMPUTER SCIENCE
COURSE CODE: COM312 **CU:** 3.0
COURSE TITLE: DATABASE DESIGN I
DURATION: 2½hrs
INSTRUCTIONS:- **ANSWER Question One & Any Other Three Questions [60 marks]**

Question one

a. Jumia Company has a delivery order database which maintains customer's records of order(s) through the Salesmen. A customer can place order(s) through single Salesman or numerous Salesmen. The data requirements are summarized as follows:

i. Jumia Delivery order company has Salesmen each identified by a unique Salesmen number, first, middle and last name, salary and several contact numbers
ii. Each customer of the company is identified by a unique customer Identification number and first name, others are address, derived age, product ordered and multiples bank account
iii. Each order placed by a customer is taken by the company Salesmen and each is given a unique order number and amount.

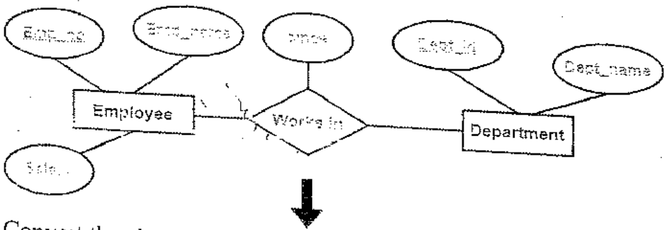
Construct an ER diagram using the above Jumia Delivery orders details [10Marks]

b. Data model can be divided into three categories according to the degree of abstractions hence, list the three categories of data model [3Marks]

c. List five (5) advantages that DBMS have over conventional file system [5Marks]

Question Two

a.



Convert the above ER diagram to relational database. [5Marks]

b. Shared file based approach was used to improve the problem of data redundancy and inconsistent data between different applications, however there are certain drawback of using this approach, list four (4) downside of shared file based approach in storing data [4 Marks]

c. List five types of data model [5Marks]

Question Three

a. Briefly explain the following terms
   I. Entity Relationship Model [2Marks]
   II. Alternative Key [2Marks]
   III. Derived Attribute: [2Marks]

b. List four database users [4Marks]

c. State four purpose of data model [4Marks]

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Costing

Question Four

Customer_name	DateOpened	Branch	AccountNo
Jacob	24/4/2011	Yola	201
Daniel	5/2/2012	Hong	355
Binta	11/7/2011	Mubi	457
Frank	15/6/2010	Numan	612
Simon	5/3/2012	Gulak	457

a. Consider the instance of the Customer **Account** table above, which lists the names of customers for DASHBANK in Adamawa State, their account number and the date the account was opened. Write the SQL queries for the below questions.

(i) Table is created under the database 'DashBank' [2Marks]
(ii) Create the table [21/2 Marks]
(iii) Insert the tuples using the first three records [21/2Marks]
(iv) Retrieve the entire records from the customer table [1Marks]
(v) Display the information of all the Customer based on branch in descending order [2Marks]

B. Distinguish between distributed and parallel computing systems. [4Marks]

Question five

. Examine the table shown below.

StudentID	Name	SemesterID	CarryOvers	Level
ST/CS/ND/20/001	Ashton	4	COM211,COM212, COM113	NDII
ST/CS/HND/21/065	Wilson	2	COM315	HNDI
ST/CS/ND/19/090	Ali	4	COM213, EED216	NDII
ST/CS/HND/20/144	Jummy	2	COM321,COM333,OTM415	HNDII

i. State the basic laws of Normalization. Hence normalize the table below to 1NF. [12Marks]
b. Enumerate two purposes of normalization. [2marks]

Question six

a.

Customer Table

Attributes	Data types
CustomerId (PK)	int
CustomerName	varchar
Address	varchar
Phone	int

Product Table

Attributes	Data types
ProductId (PK)	int
ProductName	varchar
ProductDescription	varchar
Quantity	int

a. Using the above the **Customer** and **Product** tables, Write the SQL queries for the below questions.

i. Add **Age** to the Customer table [2marks]
ii. Modify the **CustomerId data types to varchar(5)** [2marks]
iii. Rename Quantity to **DatePurchased** in the product table [2Marks]
iv. Delete the **Product** table [2Marks]
v. Describe the structure of the tables [2Marks]

b. List any 4 Database System Applications you know [4Marks]

THE FEDERAL POLYTECHNIC, MUBI

SCHOOL OF SCIENCE AND TECHNOLOGY,
DEPARTMENT OF COMPUTER SCIENCE

1st & 3rd SEMESTER EXAMINAION, 2020/2021 ACADEMIC SESSION

PROGRAMME:	HND I COMPUTER SCIENCE
COURSE CODE:	COM 315
COURSE TITLE:	INTRODUCTION TO PYTHON PROGRAMMING
DURATION:	2 ½ HOURS
INSTRUCTIONS:-	ANSWER ANY FOUR (4) QUESTIONS

QUESTION ONE:

a) Enumerate five (5) features of Python — **5 Marks**

b) Evaluate the following based on operator precedence
- i) 50 - 5*6 / 4 = 2.5
- ii) 17 // 3 = 5
- iii) 25 % 3 = 1
- iv) 6. ** 2 = 36.0

5 Marks

c) What is the output of:
print(type(12E4)) output is < class 'float' >

5 Marks

QUESTION TWO

a). What is the output of the following:
```
MyList = [1,2,3,4]
YourList = [1,2,3,4]
Print(MyList.reverse()) non
MyList == YourList false
```
3 Marks

b) Write a Python program to display classes of HND: Distinction, Upper, Lower, Pass and Fail. — **8 Marks**

c) What is the of the following:
```
i = 1
while i < 6:
  print(i)  → 1
  if i == 3:  — 9
    break  — File "<stdin>", line 1 SyntaxError: 'break' outside loop
i += 1
```
4 Marks

QUESTION THREE

a). Give output of the followings:
```
x = 10/2**6  — ~~Accepted~~
y = 12//4%3  — ~~Accepted~~
print('x != y is', x!=y)  — File "<stdin>" print(x!= ...
print('x >= y is', x>=y)  — x >= y is true
```
4 Marks

b) Write a Python program that calculates the area of a Circle after receiving the radius from the user. — **8 Marks**

c) What is the output of the following codes:
```
word = 'Congratulations'  — Accepted
word[0:7] + word[-1] + '!'  — Congrat!
```
3 Marks

(x != y is True
x >= y is True)

QUESTION FOUR

a) Differentiate between 'integer' and 'boolean' **4 Marks**

b) Write a Python code that prints 12 multiples of 3 using a range. **7 Marks**

c) Find the outcome of the following codes:

```
a = {1, 4, 3, 5, 2}
b = {3, 1, 5, 2, 4}
print(a.union (b))
```

4 Marks

QUESTION FIVE

a) Create a recursive function that calculates the Factorials of any number. **7 Marks**

b) Create a 'Person' class with method that creates .name and .age attributes **4 Marks**

c) Give the syntax of the following SQL commands used in sqlite3:

i) UPDATE TABLE

ii) INSERT INTO

4 Marks

QUESTION SIX

a) Explain the following:

i) Numpy

ii) Matplotlib

6 Marks

b) What is the output of the following codes:

```
import matplotlib.pyplot as plt
x = [21,22,23,4,5,6,77,8,9,10,31,32,33,34,35,36,37,18,49,50,100]
num_bins = 5
plt.hist(x,num_bins, facecolor = 'black'
plt.show
```

3 Marks

c) What is casting? Hence write the code which converts string to int and added. **6 Marks**

1,2,3,4.5

COM315 TEST 2

1. Idea Fiction ....is one of the greatest features in python
2. Python can run on a wide variety of hardware platforms and has the same interface on all platforms that is python is ....Portable............
3. A development Environments a combination of a text editor and a Python runtime implementation
4. Integer (int)........ is a positive or negative whole number, without decimals and of unlimited length
5. If y=2.8, >>> print(type(y)) results to ......y result is 2.8..................
6. String..... in Python are contiguous set of characters represented in the quotation marks
7. ~~Assignment~~ Operators is used to assign values to variables.
8. If x = '100' and >>> y = '-90', print x + y results to ......'100' (10)...... in float it's (10.0)
9. If MyList1= [1,2,3,4,5,5,5,5], print(MyList1.count(5)) results to ....
10. If x = 10 and y = 12, print('x != y is',x!=y) results to:......True......x != true
11. If x = 40, x = x // 5 produces ....
12. print("abc DEF".capitalize()) results to ....
13. If a=set[1, 4, 3, 5, 2] and b=set[3, 1, 5, 2, 4], print(a.union b) results to {1, 2, 3, 4, 5}.........

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page 14

14.
```
i = 1
while i < 6:
  print(i)
  if i == 3:
    break
i += 1
```

Output
1
2
3

15.
```
fruits = ["apple", "banana", "cherry"]
  for x in fruits:
   if x == "banana":
  continue
  print(x)
```

Output
apple
banana
cherry

16.
```
for x in range(6):
  if x == 3:
    break
    print(x)
  else:
  print("Finally finished!")
```

Output: 0
1
2

for break
~~Output~~ is (i) value

Different b/w Python and other programming languages

1. Python is a Special programming language which differs from the other langu... ~~its~~ it is not confined to using keywords (like "begin" and "end" as in Pascal)
2. Python does not use Special characters (like ";" as in C++).
3. Python uses '\' to indicate continuation of the previous line.
4. Python is strict on indentation, which when not obeyed will make your program err...

The Federal Polytechnic Mubi,
Department of Mathematics and Statistics,
HND 1 Computer Science ,
First Semester Examination ,2021
Course:- Statistical Method 1 (**STA 311**)
Instruction: - Answer Only five (**5**) questions. Time: - $3\frac{1}{2}$ hrs.

1 a. A continuous random variable X has the pdf f(x)= $\frac{x^3+8}{16}$; for $0 < x < 2$.
Find (i) $P(\frac{1}{2} < x < \frac{2}{3})$, (ii) $P(3 < x < 4)$

2. Given that

$$f(x) = \begin{cases} (Sx^2(3-x)) & 0 < x < 3, \\ (0) \text{ otherwise }, & \end{cases}$$

i. Find the constant S. (ii). Show that it is pdf (iii). Compute $P(2 < x < 3)$

3. The time required for Modi to travel from Maiduguri to Mubi follows a pdf $f(x) = \frac{12}{250}(5x - x^2)$; $0 < x < 5$
(a). The probability that she travel at random required between two and three
(b) Hence, determine the expected value and variance.

4. The joint density function of two random variablse X and Y is given by

$$f(x,y) = \begin{cases} (\frac{xy}{56}) & 1 < x < 2, 3 < y < 4, \\ (0) \text{ otherwise }, & \end{cases}$$

Find (i) E(X) (ii) E(Y) (iii) E(4X + 5Y)

5a. Find the value of K if f(x,y) =K(1-x)(1-y) for $x > 0, y < 1$) is be a joint density function X and Y
b. Given that

$$f(x) = \begin{cases} (13.5x - 1.35) & 0.1 < x < 0.2, \\ (0) \text{ otherwise }, & \end{cases}$$

Find $P(0.3 < x < 0.4)$

6. A perfect coin is tossed twice. Find the moment generating function of the number of tails.
b. Let X have the exponential density function. Find Moment generating function.

$$f(x) = \begin{cases} (\theta e^{-\theta x}) & x > 0, \\ (0) \text{ otherwise }, & \end{cases}$$

7. Explain the term "Mutually exclusive events". (i). A pair of dice is thrown once. If the sum on the faces is 10. Find the probability that the two faces are equal..
(ii) If the sum of the faces is 8. Find the probability that one of the faces is a 3.

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THE FEDERAL POLYTECHNIC MUBI
SCHOOL OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
1ST AND 3RD SEMESTER EXAMINATION, 2020/2021 SESSION

PROGRAMME: HND I COMPUTER SCIENCE
COURSE CODE: COM 315 CU: 4.0
COURSE TITLE: INTRODUCTION TO PYTHON PROGRAMMING
DURATION: 2 ½ HOURS
INSTRUCTIONS:- ANSWER ANY FOUR (4) QUESTIONS (60 MARKS)

Question One:

a) List 2 areas of application of Python Programming language — **4 Marks**
b) Outline the differences between Scripting and non-scripting language — **4 Marks**
c) Write a Python program to convert temperature from Fahrenheit (°F) to Celsius (°C). (Hint: °C = 5/9(°F – 32)). — **7 Marks**

Question Two:

a) List any 2 reasons why IDE is necessary in Python. — **3 Marks**
b) Enumerate the basic Python structures. — **6 Marks**
c) Write a Python program to compute the Area of a Circle after the user inputs the radius. (Hint: Area = πr^2). — **6 Marks**

Question Three:

a) Explain with examples the following data types: — **5 Marks**
   i) Float
   ii) String
b) Write the output of the following programs: — **10 Marks**
   i) print("abcdef".find("cd")) 2 cd 2
   ii) print("xyyzxyzxzxyy".count('yy')) -1 7 2
   iii) example.count('l'). If example = "hello"
   iv) print(a==b). if a = [1, 4, 3, 5, 2] and b = [3, 1, 5, 2, 4] False
   v) print(s1.issubset(s2)), if s1 = {1, 2, 3} and s2 = {1,2,3, 4, 5, 6} 123

Question Four:

a) Justify the Python's controversial choice of 'j' instead of 'i', in Complex Arithmetic. — **6 Marks**
b) What is PowerShell to Python? — **2 Marks**
c) Write a Python program to detect whether the year entered by a user is a Leap Year or not. — **7 Marks**

Question Five:

a) Explain with examples the use of the following methods: — **8 Marks**
   i) Clear() – in List
   ii) Intersection() – in Set
b) If word = 'Conventional', give the command that will output 'vent'. — **2 Marks**
c) What is the output of the following code: — **5 Marks**

```
for x in range(5, 50, 5):
print(x)
```

Question Six:

a) What is the output of the following function: — **3 Marks**

```
def children(fname, lname):
        print(fname+ " "+lname)
children("Adamu")
```

b) Write Python-Sqlite3 command for: — **6 Marks**
   i) Connecting to database 'Customer'
   ii) Committing records
c) Give the command for creating a Class named Student to inherit from a Class named People, with no properties. — **6 Marks**

THE FEDERAL POLYTECHNIC, MUBI

SCHOOL OF SCIENCE AND TECHNOLOGY,

DEPARTMENT OF COMPUTER SCIENCE

1st & 3rd SEMESTER EXAMINAION, 2020/2021 ACADEMIC SESSION

PROGRAMME: HND I COMPUTER SCIENCE
COURSE CODE: COM 314
COURSE TITLE: COMPUTER ARCHITECTURE
DURATION: $2^1/_2$ Hrs
INSTRUCTION: ANSWER FOUR (4) QUESTIONS ONLY

Q1
a. What is Digital computer organization?
b. Computer architect designs the system from building blocks that include the basic processor, **$1^1/_2$ Marks**
memories of various performance levels, input and output devices, and interconnection buses, explain. **$7^1/_2$ Marks**
c. Explain Serial bus organization and compare it with parallel bus organization. **4marks**
d. Why are Buses used in computer? **2 Marks**

Q2
a. Explain Instruction format and its various constituents **5 Marks**
b. Using a suitable diagram describe the sequence of operation and instruction format of three (3) address machine. **6 Marks**
c. List any four (4) role of Computer architect. **4 Marks**

Q3
a. Explain how the instruction A=B+C+D can be executed in digital system. **10 Marks**
b. Explain any two (2) of the following i).Throughput ii).Latency iii).Prefetching. **5Mark**

Q4
a. What is Instruction set architecture? **2 Marks**
b. Explain Fetch- Execute cycle **4 Marks**
c. Explain the working mechanism of a Zero address machine **7 Marks**
d. What is Instruction level parallelism? **2 Marks**

Q5
a. Explain pipelining and state its advantage over non – pipelining process. **6 Marks**
b. Pipelining as a modern technique used both in computer and digital electronic devices to Increase instruction throughput. Explain using example with five (5) instructions to be executed. **6 Marks**
c. Explain the functions of the instruction level IF, ID, EX, MEM and WB. **3 Marks**

Q6
a. What is a register? **2 Marks**
b. List and explain any four (4) registers you know **4 Marks**
c. Explain the term hazard in computer architecture. **2 Marks**
d. Explain data hazard with the aid of a diagram **7 Marks**

Thermistors: Thermistors (also known as [illegible] sensitive resistors') are semiconductors, made from a specific mixture of [illegible] nickel, manganese, copper, cobalt, iron magnesium, titanium and other metals [illegible] temperatures above 982°C.

THE FEDERAL POLYTECHNIC MUBI
SCHOOL OF SCIENCE AND TECHNOLOGY
THE DEPARTMENT COMPUTER SCIENCE
1st AND 3rd SEMESTER EXAMINATION, 2021/2022

PROGRAMME: HND 1 COMPUTER SCIENCE
COURSE CODE: COM 314 CU: 3.0
COURSE TITLE: COMPUTER ARCHITECTURE
DURATION: 2 Hrs
INSTRUCTION: ANSWER FOUR (4) QUESTIONS ONLY (60 marks)

Q1a. What is Digital computer organization **2 Marks**
b. Mention and explain each of the five (5) functional part of a computer **7$^1/_2$ Marks**
ci. What is a Bus **1$^1/_2$ Marks**
ii. Mention one each Advantage and Disadvantage of Serial and Parallel bus. **4 Marks**

Q2a. Explain Instruction format and its various constituents. **3 Marks**
b. Describe the sequence operation of 4 Address machine (Diagram is necessary). **8 Marks**
c. List any four (4) role of Computer architect. **4 Marks**

Q3a. Itemize the necessary steps in order to execute a set of instructions (Program). **10 Marks**
b. Explain any two (2) of the following i).Throughput ii).Latency iii).Prefetching. **2$^1/_2$ Marks**

Q4a. What is Instruction set architecture **2 Marks**
b. Explain Fetch- Execute cycle **4 Marks**
c. Explain the working mechanism of an Accumulator machines **7 Marks**
d. what is Instruction level parallelism **2 Marks**

Q5a. Explain the concept Pipelining. **3 Marks**
b. Pipelining as a modern technique used both in computer and digital electronic devices to increase instruction throughput. Explain by citing example with Six (6) instructions to be executed. **8 Marks**
c. Itemize the benefit of pipelining over non-pipelining processor **4 Marks**

Q6a. What is a register **2 Marks**
b. List and explain any four (4) register you know **4 Marks**
c. Pipelining is an implementation technique whereby multiple instruction are overlapped in execution but sometimes cease to work correctly due to some factors. Explain **9 Marks**

THE FEDERAL POLYTECHNIC, MUBI

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DEPARTMENT OF COMPUTER SCIENCE

1st & 3rd SEMESTER EXAMINAION, 2020/2021 ACADEMIC SESSION

PROGRAMME: HND 1 COMPUTER SCIENCE
COURSE CODE: COM 311
COURSE TITLE: OPERATING SYSTEM I
DURATION: $2^1/_2$ Hrs
INSTRUCTION: ANSWER FOUR (4) QUESTIONS ONLY

Q1
a. Define Operating System — 3 Marks
b. List and explain four (4) functions of Operating system — 6 Marks
c. Explain any four (4) types of Operating system — 6 Marks

Q2
a.i What is a Process in Operating system? — 2 Marks
a.ii. State four (4) difference between a Process and Program — 4 Marks
b. List and explain the components of a Process — 4 Marks
c. With the aid of a diagram explains a Process state. — 5 Marks

Q3
a. What is Process scheduling? — 1 Marks
bi. Briefly explain Process scheduling queue and its types — 6 Marks
bii. Differentiate between Process scheduling and Process scheduler — 2 Marks
c. Differentiate between Long Term, Short Term and Medium term Scheduler 6 Marks

Q4
ai. What is a Thread? — 2 Marks
ii. What is the relationship between a Process and Thread — 4 Marks
b List and explain any three of Multithreading model — 6 Marks
c List any three advantages of Multithreading model — 3 Marks

Q5
ai. What is Cooperative process? — 2 Marks
aii. State any Four (4) advantages of Process cooperation — 4 Marks
b. Differentiate between Direct and indirect communication — 3 Marks
c. Explain any three (3) of the following i). Non- preemptive scheduling ii). Preemptive scheduling iii). Context switching iv). Dispatcher — 6 Marks

Q6
a. What is Scheduling? — 1 Mark
b. Use the following details below to explain how the above processes will be Executed using the following i). Priority ii). First come First serve iii) Shortest job first. — 6 Marks

Process	Burst time (Millisecond)	Priority
P1	10	1
P2	6	5
P3	9	2
P4	4	3
P5	5	4

c. What is interrupt? — 2 Marks
d. List and explain the two (2) basic types of interrupt — 6 Marks

interrupt

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF COMPUTER SCIENCE
2^{ND} & 4^{TH} SEMESTER EXAMINATION, 2020/2021 SESSION

PROGRAMME: HND I Computer Science
COURSE: Operating System **DURATION:** $2^{1/2}$ Hrs
INSTRUCTION: Answer any five (5) questions only.

QUESTION 1

a. Briefly describe how Operating System manages memory. 5mrks
b. Enumerate any four (4) functions of memory management. 4mrks
c. Differentiate between contiguous and non-contiguous memory allocation. 5mrks

QUESTION 2

a. In a tabular form, discuss the strengths and weaknesses of the following memory management techniques:
i. Fixed partitioning ii. Dynamic partitioning iii. Simple Paging iv. Simple Segmentation 8mrks
b. List any three (3) advantages and disadvantages each of Demand Paging. 6mrks

QUESTION 3

a. With the aid of a diagram, describe a virtual memory. 10mrks
b. Virtual memory can be implemented through two methods, mention them. 2mrks
c. What is Page replacement? 2mrks

QUESTION 4

a. Using a Reference string 701203042303212017, show the following page replacement algorithm:
i. First-In-First-Out ii. Optimal Page Replacement iii. LRU page-replacement 10mrks
b. Mention the conditions for deadlock. 4mrks

QUESTION 5

a. Define a File 3mrks
b. Write on the following terms: i. A field ii. A record iii. A database 6mrks
c. Highlight any five (5) attributes of a file. 5mrks

QUESTION 6

a. Write short note on these file operations
i. Creating a file ii. Writing a file iii. Reading a file iv. Deleting a file v. Truncating a file 5mrks
b. Consider a user program of logical address of size 6 pages and page size is 4 bytes. The physical address contains 300 frames. The user program consists of 22 instructions a, b, c, . . . u, v . Each instruction takes 1 byte. Assume at that time the free frames are 7, 26, 52, 20, 55, 6, 18, 21, 70, and 90.
Find the following?
a) Draw the logical and physical maps and page tables? 9mrks
b) Allocate each page in the corresponding frame?
c) Find the physical addresses for the instructions m, d, v, r?

QUESTION 7

a. Mention and discuss the 2 major methods of accessing information. 9mrks
b. When is a process said to be deadlock? 2mrks
c. List the 3 strategies to handle deadlocks. 3mrks

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